

Green alternatives of radiation shielding glass using rice husk ash: Dose estimation in human organs

Author links open overlay

panelW. Wongwan ^{a b}, [P. Yasaka ^{a b}](#), C. Mutuwong ^{a b}, S. Yonphan ^{a b}, [W. Chaiphaksa ^{a b}](#), W. C heewasukhanont ^{c d}, T. Sareein ^e, N. Intachai ^c, [S. Kothan ^c](#), [J. Kaewkhao ^{a b}](#)

Show more

Add to Mendeley

Share

Cite

<https://doi.org/10.1016/j.anucene.2025.111797>Get rights and content

Highlights

• •

Rice husk ash (RHA) was used as a sustainable silica source for lead-free radiation shielding glass.

• •

Increasing BaO content (10–40 mol%) enhanced gamma-ray attenuation and glass density.

• •

Mass attenuation coefficients and effective atomic numbers were validated via experiments and simulations.

• •

Monte Carlo simulations with MRCP-AM phantom showed 12.66–20.41% dose reduction in human organs.

Abstract

This study presents the development of lead-free radiation shielding glass using rice husk ash (RHA) as a silica substitute. Glasses with 10–40 mol% BaO were synthesized by melt-quenching and characterized in terms of gamma-ray attenuation, elastic, and thermal properties. Increasing BaO raised the glass density from 3.236 to 3.838 g/cm³ and enhanced the mass attenuation coefficient at 0.662 MeV. The half-value layer (HVL) decreased from 2.81 to 2.37 cm, and the effective atomic number (Z_{eff}) increased to 58.2. Monte Carlo simulations showed that the BaO-RHA glass reduced organ doses by 12.66–

20.41 %, with the right lung showing the highest reduction. However, both the longitudinal and shear moduli decreased from 58.24 to 44.76 GPa and 19.18 to 11.61 GPa, respectively, indicating reduced mechanical stiffness. The glass exhibited good thermal stability and 45–85 % visible transmittance. These results demonstrate that BaO-doped RHA glass is a promising, sustainable material for radiation protection.

Introduction

In the present era, radiation has significantly advanced fields such as medical imaging, cancer treatment, and scientific research, enabling the development of technologies such as X-ray, CT scans, and radiation therapy. Despite its many benefits, ionizing radiation poses serious risks, including DNA damage and increased cancer risk (Mutuwong et al., 2024). Effective shielding materials are essential for minimizing radiation exposure and ensuring safety. Traditional shielding materials like lead and concrete are widely used for their proven ability to block radiation. Lead, known for its high density, effectively attenuates radiation, while concrete offers cost-effective bulk shielding. However, lead is toxic and poses environmental concerns, and concrete's brittleness and limited transparency restrict its applications. These challenges have prompted the search for safer and more versatile alternatives. Glass has emerged as a promising material due to its transparency, stability, and ease of fabrication (Boonin et al., 2020). Several works of literature have explored various heavy metal oxides and modified borate glasses to enhance radiation attenuation performance, such as the use of Bi_2O_3 (Naseer, 2025), CaO (Kaky et al., 2025), and combinations like $\text{PbO}:\text{B}_2\text{O}_3:\text{CuO}:\text{CaO}$ (Setiawan et al., 2024). These works support the growing trend of tailoring glass compositions for optimized shielding applications. One of the primary components of glass is silicon dioxide (SiO_2), a compound renowned for its structural integrity and ability to form durable, transparent materials. However, to reduce the cost of synthesis and environmental problems, researchers have turned to rice husk ash (RHA) as an alternative to the chemical because rice husk ash has a high purity of silica dioxide (Wahab et al., 2020). In 2023, Thailand generated 8.6 million tons of rice husk ash as a byproduct of rice production. Most of these husks are either openly burned after harvest or used for low-value purposes like fertilizer or mushroom farming, leading to a waste management challenge for rice-producing countries. In addition to using RHA instead of SiO_2 , Borate oxide (B_2O_3) was used in this work because it is a network glass with high stability, high transparency, and thermal stability (Wongwan et al., 2023). TeO_2 is one glass network that has received a lot of attention because of its high refractive index (>1.6), and low melting point ($\sim 750\text{--}950\text{ }^\circ\text{C}$) (Kumar et al., 2003). This characteristic is suitable for application in radiation shielding. Adding alkali metals to the glass host is important because they can be thermally activated, allowing them to move within the glass and easily replace other ions of the same

valence. The addition of sodium oxide to the silicate network can increase the glass transition temperature (T_g) and reduce the thermal expansion coefficient. Similarly, incorporating barium oxide (BaO) into the glass improves its ability to shield against X-rays and gamma rays, while also offering a safer alternative to lead for both human health and the environment (Tuscharoen et al., 2012). However, previous studies have only focused on the material properties and haven't assessed radiation exposure to the human body before and after using this glass. Advanced simulations like Monte Carlo methods are promising for better understanding and optimizing these materials. These simulations, when used with computational human phantoms (CHPs), can precisely model how radiation interacts with the human body, providing valuable insights into the effectiveness of the materials (Golshanian et al., 2017). CHPs are detailed 3D human body models, including organs and tissues. Monte Carlo simulations offer accurate 3D dose calculations and are efficient in terms of time and cost. The newer tetrahedral phantoms address issues found in earlier models, such as rough surfaces, low resolution in voxel phantoms, and slower speeds in polygon phantoms (Mutuwong et al., 2024, Kim et al., 2020).

In this study, glass samples with the composition (55-x) RHA: 20B₂O₃: 15TeO₂: 10Na₂O: xBaO (where x = 10, 15, 20, 25, 30, 35, 40 mol%) were prepared using the conventional melt quenching method for radiation shielding purposes. The gamma-ray shielding performance was assessed through the PHITS and WinXCom programs by examining parameters such as the mass attenuation coefficient (μ_m), half-value layer (HVL), mean free path (MFP), effective atomic number (Z_{eff}), and effective electron number (N_{eff}) within the energy range of 0.20 to 1 MeV. To validate the simulated and theoretical results, experimental techniques were employed using Compton scattering with various energy sources, including ¹³⁷Cs, for comparison. Additionally, adult male mesh-type reference computational phantoms (MRCP-AM) were used to simulate the human body. PHITS Monte Carlo simulations estimated the absorbed dose and effective dose rate for the phantom, both with and without 1.0 cm thick shielding glass, using a mono-energetic photon of 0.662 MeV in the anterior-posterior (AP) direction. Therefore, this study aims to evaluate the gamma shielding and organ-specific dose reduction capabilities of BaO-doped RHA glass using both experimental and simulation techniques.

Gamma-ray shielding parameters

Measurement of the linear attenuation coefficient (μ) and mass attenuation coefficients (μ_m) of radiation shielding materials can be calculated by Lambert-Beer law (Mutuwong et al., 2024).

In this context, I_0 represents the intensity of photons before passing through the sample, I is the intensity of photons after passing through the sample, d , is the sample's thickness, and ρ is the sample's density.

The error in the photon energy (E_r) represents the uncertainty in

Glass preparation and characterization

In this experiment, RHA was obtained from Lop Buri Province, Thailand. The RHA was burned at temperatures of 400, 600, 800, 1000, and 1100 °C for 10 h. The composition was analyzed using X-ray fluorescence (XRF) spectroscopy (Minipal-4) supplied by Panalytical to find a suitable composition for preparing glass samples as shown on Table 1. The increase in temperature leads to a higher weight percentage (wt%) of SiO_2 in rice husk ash (RHA). This is because higher temperatures effectively remove

Density, and molar volume

The (55-x) RHA: $20\text{B}_2\text{O}_3$: 15TeO_2 : $10\text{Na}_2\text{O}$: $x\text{BaO}$ glasses (where $x = 10, 15, 20, 25, 30, 35$, and 40 mol%) were polished to a thickness of 0.2 cm to measure various properties as shown in Fig. 4. The glass densities were measured using Archimedes' principle from Equation (15). Where W_{air} is the weight of glass in the air (g), W_{water} is the weight of glass in distilled water (g), and ρ_{water} is the density of distilled water, which is about 1 g/cm³. The increase in BaO

Conclusions

This study successfully developed a lead-free radiation shielding glass by incorporating BaO into a RHA-based glass system. The substitution of SiO_2 with BaO increased the glass density and improved gamma-ray attenuation, as reflected by enhanced μ_m , Z_{eff} , and reduced HVL values. PHITS Monte Carlo simulations confirmed that the use of BaO-doped glass reduced organ-specific radiation doses by 12.66–20.41 %, with the right lung exhibiting the greatest reduction. Although mechanical stiffness

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

5. Acknowledgments

This research has received funding support from the NSRF via the Program Management Unit for Human Resources & Institutional Development, Research and Innovation [grant number B13F670061]. Additionally, this research has received funding support from the NSRF via the Program Management Unit for Human Resources & Institutional

Development, Research and Innovation [grant number B13F680084]. We would like to express our gratitude to Thailand Science Research and Innovation and Nakhon Pathom